When Better Retrieval Makes Worse Answers: Downstream-Gated Lifecycle Memory Across Eighteen Research Directions

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Abstract

Long-running programming agents must preserve decisions while avoiding constraints that were renamed, corrected, reverted, deleted, or made invalid by a new branch or environment. Static retrieval can surface obsolete statements; deleting them discards provenance. We study a narrower question above an existing memory engine: how can a bounded lifecycle sidecar propose policy updates from feedback while preventing a retrieval proxy from promoting a worse answer policy?

We implement correction-linked predecessor–successor redirection, fixed-cost fallback, protocol-versioned policy search, simple temporal baselines, single-factor ablations, proxy-alignment diagnostics, and text-only correction detection in MemoryCore. The public experiments use Memora, MemOps, and LongMemEval-V2 as method-validation substrates rather than evidence about private programming traffic. On a frozen 50-question panel conditioned on obsolete Base exposure, fixed-budget V1 raises answer FAMA by 8.09 points (95% persona-cluster CI: 5.17–12.59). A new 200-question census that is not conditioned on Base failure finds a smaller full-population gain of 1.42 points (CI: −0.005–3.08 points), but a confirmed 2.96-point gain on 92 forgetting-bearing questions and a remembering-MPA harm signal. Thus V1 is a conditional incumbent, not a universal default.

A proxy-selected V2 policy improves direct retrieval but loses 4.08 FAA points and uses 37.5% more context than V1 on the conditioned panel. On the broad panel, none of four simple baselines, six isolated V2 factors, or combined V2 passes the frozen promotion gate. A stance-aware proxy raises Kendall's τ_b with answer FAMA only from 0.529 to 0.538. A text-only detector identifies invalidating sessions at 97.89% precision and 79.64% recall on persona-disjoint test histories, yet predecessor-link precision is only 20.05%; this blocks an end-to-end correction-construction claim. Separately, evidence-preserving compression reduces injected tokens by 20.84% on a checksum-frozen LongMemEval-V2 test while direct answer-atom support is unchanged, but no answer-level noninferiority is established.

Across eighteen directions, the durable result is a separation of proposal, promotion, and evidence scope: cheap feedback may propose a policy, downstream answers and cost must promote it, and uncertain correction links must abstain. Exact operational fallback is necessary but not semantic safety. The resulting code and adapters are portable to programming sessions, while repository-, branch-, commit-, and environment-scoped validity remains future work.

1 Introduction

Interactive software tasks evolve. A user can replace REST with GraphQL, rename an interface, invalidate a test expectation, revert a design, switch branches, or change the toolchain. A memory system that retrieves both an old and a current constraint can confidently reintroduce a rejected decision. Lexical and embedding relevance do not imply current validity. Permanently deleting history is also undesirable: provenance is lost, historical questions become harder, and the correction that explains current state may itself be inaccessible.

Long-term memory systems address hierarchy, extraction, consolidation, graph structure, and context management [8, 15, 1, 9]. Benchmarks reveal failures in temporal reasoning, updating, and selective forgetting [6, 14, 4]. Memora explicitly separates required memory from invalidated memory through Memory Presence Accuracy (MPA), Forgetting Absence Accuracy (FAA), and Forgetting-Aware Memory Accuracy (FAMA) [13]. MEMPROBE further argues that task completion and the recoverable state retained by memory are distinct capabilities [5].

This paper studies lifecycle control above MemoryCore, an open-source hierarchical memory service with conversation retrieval, atomic memories, scenes, profiles, HTTP access, SQLite FTS5, and vector retrieval [12]. We do not retrain the extractor or replace the base index. Instead, a bounded sidecar records correction relations, transforms a fixed candidate list, and returns the exact base call on disablement or failure.

The central engineering risk is optimization by an incomplete proxy. A direct audit may show more current evidence and less obsolete evidence, while the generated answer becomes worse because context wording, entity anchors, extra slots, or decoder behavior changes. We therefore separate a *candidate proposer* from a *promotion gate*. The proposer may use cheap feedback; promotion requires frozen downstream quality, forgetting, cost, uncertainty, and integrity checks.

Our contributions are:

- a correction-linked, fixed-budget lifecycle sidecar with bounded traversal, capacity, latency, switch-off, and whole-call BASE fallback;
- a proposal–promotion protocol that rejects a direct-proxy winner after answer generation, and distinguishes quality promotion from efficiency promotion;
- a broad 200-question, 13-arm answer experiment with strong simple baselines and single-factor ablations, using MiniMax-M3 and DeepSeek-V4-Flash in a crossed no-self-judgment design;
- an eighteen-direction research ledger that retains negative results in shielding, routing, state projection, transition memory, procedure memory, evidence expansion, structured refutation, proxy design, and automatic linking;
- a text-only, persona-disjoint correction-construction audit showing that event detection transfers but predecessor linking does not;
- a separate evidence-preserving compression result and an explicit portability boundary for repository-, branch-, commit-, worktree-, and environment-scoped programming memory.

The main conclusion is deliberately bounded. Correction-linked management is useful when obsolete memory is present and the relation is known. The evidence does not justify enabling V1 on all traffic, does not validate private programming tasks, and does not solve end-to-end correction linking.

2 Related Work and Research Position

Long-term agent memory. MemGPT treats context management like a virtual-memory hierarchy [8]. Mem0 extracts and consolidates persistent information, while A-MEM builds an evolving linked note network [1, 15]. Zep and its open-source Graphiti engine attach validity windows and provenance to evolving facts [9, 19]. Our mechanism is narrower: it augments, rather than replaces, an existing retriever and constrains every query-time action.

Evaluation beyond recall. LoCoMo and LongMemEval expose long-session temporal and update failures [6, 14]. MemoryAgentBench identifies accurate retrieval, test-time learning, long-range understanding, and selective forgetting as separate competencies [4]. Memora evaluates remembering, reasoning, and recommending over weeks to months and penalizes invalid-memory reuse [13]. MEMPROBE directly audits the retained memory artifact and reports that assistance can saturate while state recovery remains moderate [5]. These works motivate our separation of direct memory evidence from answer-level use.

Feedback and self-optimization. Reflexion updates agent behavior with verbal feedback [11]. Memory-R1 learns add/update/delete/no-op memory operations, SelfMem refines memory strategy from environmental feedback, and DeferMem learns query-time evidence selection [16, 17, 18]. We do not reproduce their training procedures. Our focus is a reviewable, bounded policy space and an incumbent-preserving promotion protocol for an existing store.

Memory for coding agents. SWE-MeM lets an agent decide when and how to compress long software-engineering trajectories and jointly optimizes memory with issue resolution [3]. Structurally aligned subtask memory reports that monolithic instance memory suffers from a granularity mismatch [10]. ToM-SWE maintains persistent user goals and constraints across stateful software-engineering interactions [20]. These studies demonstrate the relevance of memory to coding agents, but they do not remove our public-data limitation: Memora has no repository, branch, commit, worktree, or toolchain labels. We therefore present programming transfer as a scoped adapter design, not an empirical result.

3 Problem Formulation

Let the unchanged base retriever return a ranked pool

$$B(q) = (m_1, \ldots, m_K),$$

and let its injected prefix be $B_k(q)$. A lifecycle event is

$$e = (t_e, \kappa_e, c_e, V_e^-, S_e, s_e),$$

where t_e is sequence, $\kappa_e \in \{\text{update}, \text{delete}\}$ is kind, c_e is confidence, V_e^- contains invalidated values, S_e contains successor memory identifiers, and s_e records provenance. A predecessor memory receives an edge to S_e when it predates the event and contains an invalidated value.

A policy π_θ maps $B(q)$ and the bounded lifecycle ledger to an injected list. The policy must satisfy capacity, latency, result-count, and integrity bounds. If the sidecar is disabled, malformed, cyclic, over capacity, or timed out, the entire call returns $B_k(q)$ rather than a partial transformation.

We distinguish three objectives:

1. **quality**: improve downstream answers without violating forgetting safety or cost;
2. **efficiency**: preserve a frozen quality invariant while reducing context cost;
3. **construction**: infer lifecycle events and predecessor links without oracle operation metadata.

Changing the objective after scoring is forbidden. In particular, a candidate frozen as a quality improvement cannot be retroactively promoted because it saved tokens.

4 Lifecycle Policies and Promotion

4.1 Base and fixed-budget V1

Base is the unmodified MemoryCore L0 retrieval and injection path. The research incumbent V1 follows the latest eligible correction edge for an obsolete base candidate, removes duplicate leaves, and refills from the same base pool. Its fixed parameters are

$$\tau = 0.85, \qquad H = 1, \qquad E = 64, \qquad k = 5, \qquad T = 10 \text{ ms},$$

for confidence, hop count, expansion capacity, output count, and sidecar wall time. The ledger may link a predecessor directly to the latest eligible correction, so one query-time hop can represent a longer update history.

The Memora implementation constructs V1 edges from released write-time operation fields. These fields do not contain evaluation answers, but they do make the management experiment quasi-oracle with respect to event type and invalidated value. Section 10 separately evaluates a text-only constructor.

4.2 Contextual V2

The V2 candidate space varies confidence $\tau \in \{.85, .91, .96\}$, hops $H \in \{1, 2\}$, redirect-conditioned extra slots $s \in \{0, 1, 2\}$, and historical-aggregate protection $p \in \{0, 1\}$. The 36-policy proposer optimizes a direct forgetting-aware proxy on weekly/monthly feedback and selects $\tau = .96$, $H = 2$, $s = 2$, and $p = 1$. Capacity remains 64 and time remains 10 ms. V2 is a proposal, not an automatic deployment choice.

4.3 Tiered promotion

For incumbent θ_I, challenger θ_C, and frozen checks $g_1, \ldots, g_J$, the research policy is

$$\theta^* = \begin{cases} \theta_C, & \bigwedge_{j=1}^{J} g_j = 1, \\ \theta_I, & \text{otherwise.} \end{cases}$$

Cheap feasibility checks cover direct evidence, protected-slice harm, context cost, latency, capacity, disabled equivalence, and forced failure. Only admitted candidates receive answer generation. Answer promotion covers FAMA magnitude and uncertainty, FAA, MPA, reader-specific direction, and cost. Every failed check is logged.

This mechanism is *self-optimization* within a bounded policy space: controlled feedback changes candidate parameters or auxiliary rules, while frozen downstream feedback decides whether the system state changes. It is not unconstrained self-modification.

4.4 Operational versus semantic fallback

Exact Base fallback is operational safety: a broken sidecar does not break the base service. It is not semantic safety. Base may still retrieve obsolete information. We test exact fallback extensively but never interpret it as correctness.

5 Experimental Protocol

5.1 Public datasets and adapters

Memora revision `a6493188efc836d6511ed5e4163fe3ba87da30ff` contains 600 questions, 27,614 session files, 10 personas, and weekly, monthly, and quarterly horizons. The data-manifest SHA-256 is `dfc82711...331bfc`. Only shared-memory turns enter the L0-shaped index. Memora supports answer-level current/obsolete criteria but lacks programming scope.

MemOps provides 403 paired instances, 2,558 operations, and 2,006 longitudinal probes with explicit targets. It is used for oracle state-management diagnostics. LongMemEval-V2 revision `2cc8c540` contributes 200 selected small-tier questions and 1,870 text trajectories for state-transition, procedure, and compression experiments. Dataset readers implement replaceable task, event, provenance, and metric interfaces; the main policy code does not depend on one public schema.

5.2 Metrics and aggregation

For one judged answer, MPA scores required valid facts and FAA scores avoidance of invalid facts. Following Memora, we compute

$$\mathrm{FAMA}_a = \max\left(0, \mathrm{MPA}_a - \lambda_a(1 - \mathrm{FAA}_a)\right), \qquad \lambda_a = \frac{N_{f,a}}{N_{p,a} + N_{f,a}}.$$

The aggregation order is explicit: compute MPA, FAA, and FAMA for each judged answer; average the MiniMax-reader/DeepSeek-judge and DeepSeek-reader/MiniMax-judge cells for each case–arm; then take the arithmetic mean over the frozen case population. We do not apply the nonlinear FAMA formula after separately averaging population MPA and FAA.

The direct evidence proxy requires current-session provenance plus normalized value occurrence and penalizes obsolete exposure. It is deterministic and cheap, but cannot determine whether an answer affirms, negates, or merely narrates an old value. All primary intervals are paired persona-cluster bootstrap intervals with frozen seeds.

5.3 Model evaluation

MiniMax-M3 and DeepSeek-V4-Flash [7, 2] are the current readers and judges. The conditioned panel generates each answer once and records both judges; the primary estimate excludes self-judgment. D16 uses a stricter assignment: every MiniMax answer is judged only by DeepSeek, and every DeepSeek answer only by MiniMax. Thinking is disabled and exact model IDs, official endpoints, sampling, retry semantics, and unclear-verdict handling are frozen. Agreement between model judges is not human correctness.

5.4 Chronology and data use

The original 50-case panel is conditioned on a visible Base failure: among 192 quarterly forgetting-bearing questions, 61 expose an obsolete Base Top-5 value, and five cases per persona are selected by seeded hash. It contains 43 recommending and seven remembering questions. It estimates mechanism efficacy under stale exposure, not population impact.

D16 takes all remaining 200 quarterly questions unused by either earlier 50-case answer panel. It is a census of those remaining cases and does not condition on arm outcomes or Base stale exposure. Nevertheless, quarterly Memora had been observed during earlier proxy development, so D16 is not a pristine external-dataset holdout. Its purpose is broad robustness and causal policy-factor comparison.

5.5 Work and call volume

The lifecycle benchmark contains 54 versioned protocol JSON files, 251 TypeScript source/test files (50 test files), and machine-readable raw, summary, decision, admission, and independent-validation artifacts. Excluding a legacy GPT-4o-mini secondary run and avoiding double-counting byte-identical reused cells, the current MiniMax/DeepSeek studies make 2,456 reader calls and 3,018 judge calls (5,474 total). D16 alone consumes 2,805,144 API tokens.

6 Research Trajectory Before the Broad Experiment

Table 1 summarizes D1–D15. These are retained negative and near-miss results, not a sequence in which every later number is better.

ID	Hypothesis	Main result	Decision
D1	Obsolete-value shielding	Context exposure falls 74% to 8%, but answer MPA falls 2.02 points with a negative CI.	Reject answer safety
D2	Query risk routing	Cross-fitted routing gains 2.83 FAA points but loses 1.15 MPA points; one reader's FAMA direction is negative.	Reject
D3	Delete vacancy refill	Direct proxy gains 1.40 points, below the frozen 2-point gate; tokens rise 2.89%.	Reject proxy/cost
D4	Valid-state packing	Fresh 50-case FAMA +1.16 points and tokens −3.27%; CI is [−0.14, +2.90] points.	Near miss, reject uncertainty
D5	Target-keyed state	Untouched-test state-FAMA +6.46 points, CI [−2.40, +15.10]; four Update cases harmed.	Reject test
D6	Incumbent guard	Post-hoc consumed audit: state-FAMA +6.11 points, 28 improved, zero harmed.	Viability only
D7	Transition diffs	Development support +8.13 points; validation reverses to −6.67 and tokens +3.50%.	Reject validation
D8	Learned utility/null action	Untouched-test tokens −2.92% but support changes exactly zero with no improved case.	Reject quality
D9	Outcome-gated procedures	Outcome-agnostic ablation is better on development; validation support is −11.11 points.	Reject
D10	Locally verified substitution	Tokens −33.55%, but one decisive body-text answer is removed; local feedback gives no quality lift.	Reject safety
D11	Evidence-preserving substitution	Untouched-test tokens −20.84%; 12 direct-support questions all equal to Base.	Quality reject; retain efficiency result
D12	External evidence expansion	Development direct support −3.33 points versus Base with five harms.	Reject development
D13	Residual patch	All 23 direct questions equal Base while tokens rise 0.99%.	Reject no effect
D14	Premise refutation	Two triggers among 33 premise cases, zero control false triggers; four changed reader pairs all equal Base.	Reject answer no effect
D15	Typed refutation contract	Three of four consumed design pairs improve, but fresh validation has zero triggers over 23 premise cases.	Reject zero coverage

Table 1: Research directions D1–D15. Development results are never relabeled as confirmation.

The trajectory identifies recurring mechanisms. Surface shielding removes both harmful old values and useful entity anchors. Fixed item count is not a token guarantee. Explicit target state is stronger than value-substring matching but can discard an incumbent successor. Whole-trajectory success is a noisy credit signal: failed trajectories contain useful prefixes and successful trajectories contain detours. Procedure skeletons can compress actions while losing body text, code, or diagnostics. Finally, a high-quality typed renderer cannot compensate for a trigger with zero validation coverage.

7 Conditioned V1/V2 Answer Experiment

Table 2 reports the original 50-case panel. V1 improves every primary quality metric over Base. V2 raises MPA slightly above V1 but lowers FAA by 4.08 points and uses 37.53% more injected tokens. Its FAMA difference from V1 is negative with a CI crossing zero. The gate therefore retains V1.

Metric	Base	V1	V2	V1 −Base [95% CI]	V2 −Base [95% CI]	V2 −V1 [95% CI]
MPA	0.3855	0.4342	0.4461	+0.0487 [0.0196, 0.0928]	+0.0606 [0.0216, 0.1090]	+0.0119 [−0.0039, 0.0276]
FAA	0.8531	**0.9369**	0.8961	+0.0838 [0.0599, 0.1067]	+0.0431 [0.0029, 0.0848]	−0.0408 [−0.0762,−0.0064]
FAMA	0.3195	**0.4004**	0.3961	**+0.0809 [0.0517, 0.1259]**	+0.0766 [0.0441, 0.1141]	−0.0043 [−0.0206,0.0109]
Criterion acc.	0.7040	**0.7606**	0.7382	+0.0566 [0.0385, 0.0758]	+0.0342 [0.0118, 0.0604]	−0.0224 [−0.0424,−0.0042]
Mean tokens	144.52	146.76	201.84	+1.55%	+39.66%	+37.53%

Table 2: Conditioned panel: 50 Base-stale-exposed questions, 43 recommending and seven remembering.

The conditioned result establishes that correction-linked redirection can help when obsolete evidence is present. It cannot estimate an unconditional deployment effect because selection requires a Base failure.

8 D16: Broad Baselines and Single-Factor Ablation

8.1 Design and integrity

D16 freezes 200 cases across ten personas: 100 reasoning, 62 remembering, and 38 recommending questions; 92 contain forgetting criteria and only six are Base-stale-exposed. Thirteen arms include Base, V1, four simple baselines, six isolated V2 factors, and combined V2:

- tombstone refill and latest-write-wins;
- pure recency Top-5 and explicit superseded/history rendering;
- V1 with two hops, one extra slot, two extra slots, threshold .91, threshold .96, or aggregate protection;
- combined V2.

The design contains 2,600 case–arm contexts. Reader message arrays that are byte-identical within one case are generated and judged once, then reused. This yields 947 unique prompts per reader and 1,894 reader plus 1,894 crossed-judge calls instead of 5,200 reader calls. There are zero retries, reader/judge model mismatches, or self-judgments. Two unclear verdicts are counted incorrect. An independent parser rescored all 1,894 raw evaluations, reconstructed 13 arm aggregates and 12 comparisons against V1, and found zero mismatch.

8.2 Full-population results

Table 3 shows the broad result. No challenger passes every frozen effect-size, uncertainty, FAA/MPA, cost, and integrity check.

Arm	MPA	FAA	FAMA	FAMA−V1 [95% CI]	Tokens	Token Δ	Decision
Base	0.1500	0.9780	0.1428	−0.0142 [−0.0308,0.00005]	134.19	+0.77%	baseline
V1	0.1620	**0.9881**	0.1570	–	133.16	–	incumbent
Tombstone refill	0.1601	0.9838	0.1533	−0.0037 [−0.0149,0.0070]	133.39	+0.17%	reject
Latest-write-wins	0.1598	0.9880	0.1543	−0.0027 [−0.0094,0.0039]	135.03	+1.40%	reject
Recency Top-5	0.1453	0.9821	0.1393	−0.0176 [−0.0403,0.0057]	118.97	−10.66%	reject quality
Superseded rendering	0.1559	0.9849	0.1497	−0.0073 [−0.0230,0.0080]	146.99	+10.39%	reject
V1 + H2	0.1588	0.9879	0.1536	−0.0034 [−0.0086,0]	134.42	+0.95%	reject
V1 + slot 1	0.1670	0.9862	0.1609	+0.0039 [−0.0068,0.0144]	148.66	+11.64%	reject cost/CI
V1 + slot 2	**0.1740**	0.9831	**0.1663**	+0.0093 [−0.0027,0.0227]	160.04	+20.19%	reject cost/CI
V1 $\tau = .91$	0.1616	0.9889	0.1565	−0.0004 [−0.0013,0]	133.23	+0.06%	reject
V1 $\tau = .96$	0.1656	0.9881	0.1600	+0.0030 [−0.0010,0.0083]	133.56	+0.30%	near miss
V1 + aggregate guard	0.1595	0.9881	0.1545	−0.0025 [−0.0100,0.0050]	134.14	+0.74%	reject
V2	0.1680	0.9831	0.1591	+0.0021 [−0.0073,0.0108]	147.99	+11.14%	reject

Table 3: D16 full-population crossed answer results. Token deltas are relative to V1.

The extra-slot factors reveal the central trade-off. More context increases MPA and point-estimate FAMA but reduces FAA, costs 11.6–20.2% more tokens, and does not beat V1 with confidence. Threshold .96 is the lowest-cost positive near miss; its +0.30 FAMA-point interval against V1 crosses zero. Two hops and aggregate protection are slightly harmful on this population. The simple baselines do not dominate the lifecycle incumbent.

8.3 Heterogeneous V1 effect

The full-population V1–Base FAMA difference is +0.01416 with CI $[-0.000046, +0.03082]$. FAA improves +0.01007 with a positive interval, but the primary FAMA uncertainty rule is not met. Table 4 explains why the conditioned panel and broad panel differ.

Slice	n	FAMA Δ [95% CI]	MPA Δ [95% CI]	Interpretation
Full population	200	+0.0142 [−0.00005,0.0308]	+0.0120 [−0.0031,0.0285]	positive, uncertain
Forgetting-bearing	92	**+0.0296 [0.0074,0.0549]**	**+0.0262 [0.0044,0.0517]**	confirmed positive slice
Base stale exposed	6	+0.3919 [0.2897,0.4694]	+0.3111 [0.1400,0.4286]	large, only five clusters
Recommending	38	**+0.0768 [0.0203,0.1395]**	+0.0751	strong positive task slice
Remembering	62	−0.0095 [−0.0247,0.0029]	−0.0153 [−0.0319,−0.0029]	MPA harm signal
Reasoning	100	+0.0050 [−0.0100,0.0200]	–	no confirmed effect

Table 4: D16 V1 versus Base by frozen descriptive slice. Subgroups do not override the primary gate.

The original 50-case panel oversamples exactly the setting where lifecycle correction should matter. D16 does not invalidate that mechanism result; it limits its deployment interpretation. A future selective router must use runtime-observable risk features and fresh confirmation. Benchmark task labels cannot be copied into production routing.

9 D17: Does Stance Repair the Proxy?

The released direct proxy penalizes normalized obsolete-value occurrence without distinguishing affirmation, negation, or historical narration. D17 freezes an English marker-based stance proxy: affirmed, negated, and historical mentions receive exposure weights 1.0, 0.25, and 0.1. It ranks all 13 D16 policies without reading answer outcomes for the weights.

Kendall's τ_b with answer FAMA rises from 0.5290 to 0.5385, an improvement of 0.0094 against a frozen 0.10 requirement. Correlation with MPA rises from 0.4774 to 0.4872 and with FAA from 0 to 0.0129. Pairwise direction agreement against V1 remains 8/12. Both proxies correctly predict the observed V2–V1 FAMA direction, but the overall gate fails.

This negative result rules out a tempting shortcut. Simple stance markers improve a few descriptions but do not substitute for semantic or answer-level evaluation. A learned proxy would require human calibration and still needs a downstream promotion boundary.

10 D18: Text-Only Correction Construction

10.1 Protocol

D18 directly addresses the quasi-oracle edge construction in V1. Runtime inputs are restricted to session id, persona, shared conversation text, and at most 512 earlier shared units from the same persona. Session type, operation, operation details, evaluation questions, and evidence labels are forbidden. A frozen deterministic detector recognizes explicit removal, no-longer, completion, used-to/now, from/to, preference shift, and replacement cues. A weighted lexical linker emits at most three predecessor ids and abstains when no prior unit passes its threshold.

Quarterly histories are deduplicated from weekly/monthly prefixes and split by persona: four development, three validation, and three test personas. Released operations are attached only after prediction to define invalidating sessions, kinds, obsolete values, and predecessor units. No answer-model calls are authorized.

10.2 Results

The test split contains 5,954 sessions and 1,223 gold invalidating sessions. Table 5 compares a literal remove/delete baseline, cue-only classification, and high-confidence linked prediction.

Metric	Remove/delete only	Cue only	Linked high confidence
Event precision	1.0000	0.9781	0.9789
Event recall	0.1562	0.8029	0.7964
Event F1	0.2702	0.8819	0.8783
Kind accuracy	1.0000	0.8228	0.8234
Predicted predecessor links	0	0	2,823
Link precision	–	–	0.2005
Link recall	–	–	0.1618
Any-correct-link rate	–	–	0.4538
Detector p95	–	–	5.51 ms

Table 5: D18 persona-disjoint test result. Any-correct-link rate is over true-positive linked sessions.

Event precision, recall, link recall, latency, and forbidden-read checks pass. Link precision, any-correct-link rate, and kind accuracy fail. The result is stable in direction on validation, where link precision is 19.80% and any-correct-link rate 45.74%.

The core error is entity and state ambiguity. A cue-bearing message and many prior units can share words such as "budget", "project", or "music" without referring to the same field or state. Detecting that a correction occurred is substantially easier than deciding exactly what it invalidates. We therefore classify all V1 answer results as oracle-management evidence and reject an end-to-end construction claim.

11 D11: Separate Efficiency Evidence

D11 replaces selected same-trajectory Base items with one bounded capsule only if it preserves every extracted semantic leaf, source order, provenance, unrelated Base item, item count, and token certificate. Otherwise it returns exact Base. The checksum-frozen LongMemEval-V2 test contains 30 questions, of which 12 have directly computable answer-atom support.

Across all 30 questions, D11 accepts 19 substitutions and reduces mean injected tokens from 2,796.8 to 2,213.9, a 20.84% reduction. On the 12 direct questions, Base and D11 both score 0.2083; zero improve, twelve are equal, and zero are harmed. Query p95 moves from 227.5 to 231.1 ms; selection p95 is 3.08 ms. Token, coverage, provenance, capacity, and forced-fallback violations are zero.

The frozen objective required a quality improvement, so D11 is correctly rejected for quality promotion and no answer generation is admitted. Equal direct support is not answer-level noninferiority. We retain the result only as a measured support–cost Pareto point and propose a new efficiency-specific protocol with an answer noninferiority margin.

12 Discussion

12.1 What actually works

Correction-linked redirection can improve downstream answers when the base context contains invalidated memory. It is fixed-budget, bounded, auditable, switchable, and compatible with an existing retriever. The stronger general result is the promotion architecture: V2, D1, D4, and D17 each show how a plausible or positive direct signal can fail an answer, uncertainty, or cost boundary.

The broad data suggest a risk–coverage policy rather than unconditional lifecycle rewriting. Forgetting-bearing and recommending slices benefit, while remembering MPA can be harmed. This does not yet define a deployable router: subgroup labels are diagnostic and the routing hypothesis was formed after D16 outcomes. Fresh data are required.

12.2 What did not work

More slots trade recall against obsolete exposure and cost. More hops are not automatically helpful. Recency saves tokens but loses quality. Explicitly rendering superseded facts costs more and does not dominate. Whole-trajectory outcome labels are too coarse for procedural memory. Preserving action structure without semantic leaves is unsafe. Typed refutation cannot transfer when its trigger has zero coverage. English stance markers do not align the proxy enough. Finally, lexical event-to-predecessor linking is unsafe despite strong event classification.

12.3 Transfer to programming sessions

The portable interfaces are event, scope, candidate, provenance, feedback, metric, and report adapters. An internal programming adapter should represent validity over a composite key such as

$$(\text{repo}, \text{branch}, \text{base commit}, \text{worktree}, \text{environment}, \text{task phase}, \text{file/symbol}, \text{claim type}).$$

Each claim should be active, superseded, or unknown. Unknown scope must abstain rather than invalidate.

Promising unvalidated extensions include executable witnesses (test command, exit code, commit, file hash, environment), implicit candidates from Git diff/rename/revert and test-state changes, memory rebase after branch or commit transitions, and contamination-differential evaluation that checks whether injected memory changes plans, edits, tests, or patches. A public or internal "RepoDriftBench" adapter could expose these fields without changing the lifecycle controller. These are designs, not results in this paper.

12.4 Cost and confidence

D16 demonstrates a practical two-fidelity workflow. Exact context metrics and prompt reuse cheaply eliminate many candidates; answer calls are reserved for frozen arms and unique prompts. This saved 63.6% of otherwise required reader calls in D16. Yet two model judges may share biases. The next evidence priority is a human-calibrated subset, not another larger proxy search.

13 Limitations

First, Memora is personalized dialogue rather than software engineering. It lacks repository, branch, commit, worktree, environment, component, and test scope. Second, all quarterly questions were visible to earlier proxy analysis; D16 cases are answer-unconsumed but not a pristine dataset holdout. Third, the original 50-case panel is deliberately conditioned on stale exposure and cannot estimate total-traffic impact. Fourth, model-judge agreement does not establish human correctness. Fifth, D11 has only 12 direct-support questions and no generated-answer noninferiority test. Sixth, D18 uses deterministic English cues and operation-derived scoring; it does not cover implicit code-state invalidation. Seventh, exact Base fallback protects availability, not semantic correctness. Finally, D1–D18 involve repeated scientific exploration. We preserve protocol chronology and negative results, but only a fresh external or internal dataset can support deployment claims.

14 Reproducibility and Artifacts

Every main experiment records dataset revision or manifest, protocol version, selection or census rule, seed, model configuration, raw JSONL, summary, decision, cost, fallback checks, and claim boundary. D16 provides a raw-evaluation SHA-256, independent reconstruction, and zero mismatch report. D18 provides frozen persona splits, forbidden runtime fields, per-session predictions, and artifact hashes. The structured direction registry distinguishes passed, rejected, near-miss, post-hoc, and efficiency-only outcomes.

The implementation remains a sidecar. Disabling it, corrupting its structure, forcing timeout, exceeding capacity, or encountering a cycle returns the original base call. Dataset adapters can be replaced without rewriting policy and reporting logic.

15 Conclusion

We studied memory self-optimization as controlled lifecycle policy evolution rather than unconstrained rewriting of a memory engine. Fixed-budget correction-linked management produces large gains when obsolete Base evidence is present and confirmed gains on forgetting-bearing broad cases, but it is not supported as an unconditional default. A richer proxy-selected policy, simple temporal baselines, isolated factors, and a stance-aware proxy all fail frozen promotion. Text-only correction detection is accurate, but automatic predecessor linking is not. Evidence-preserving compression reduces measured context cost but still needs answer-level noninferiority.

The final strategy is conservative: retain Base, activate lifecycle management only under high-confidence scoped evidence and a freshly validated risk gate, promote quality and efficiency on separate tracks, abstain on ambiguous links, and preserve exact operational fallback. The next substantive research problem is scoped, witness-backed invalidation for repository state—not a larger Top-k.

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